EXPERIMENT 07

Integrating a Zoho Creator Application with Another Zoho Service Using Connections and API

Library Book Issue Management — Zoho Calendar Integration Use Case

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1. Aim

To integrate the Library Book Issue Management application built in Zoho Creator with Zoho Calendar — by creating an authorised connection and calling the Zoho Calendar API through a Deluge script in a workflow — so that an event is added to the calendar the moment a member submits the library management form.

This removes the need for library staff to enter every book issue request into the calendar by hand, keeping the schedule consistent with the data captured in the application.

2. Procedure

Step 1: Open Microservices — From the Zoho Creator workspace, the Microservices page was opened. Microservices are modular components used to connect with third-party apps; Create New was selected.

Step 2: Choose Connections — In the Create New Microservice dialog (AI Models, AR Library, Connections, Datasources, Custom API), Connections — "Connect apps with third-party services" — was selected using Pick & Create.

Step 3: Pick the Zoho Calendar service — From the list of available Zoho services, Zoho Calendar was chosen as the service to connect to.

Step 4: Configure the Zoho Calendar connection — In the Zoho Calendar – Add Connection form, the Connection Name was set to Zoho Calendar Connection, Authorization Access to Administrators only, Connection Mode to Quick connect (Zoho credential), and the credentials were enabled for All Actions. The connection was then submitted with Create And Authorize.

Step 5: Authorise Creator — On the consent screen, Creator requested access to Calendar operations (add calendar, add event, get calendar and event details, free/busy, search events, update calendar and event). The consent checkbox was ticked and Accept was clicked.

Step 6: Enable the connection for the application — In the connection's Connection Access panel the link name was zoho_calendar_connection, with permissions Create Calendar, Create Event and Get all Calendar and the status Enabled. Access was switched On for Library Book Issue Management and left Off for My first Creator app.

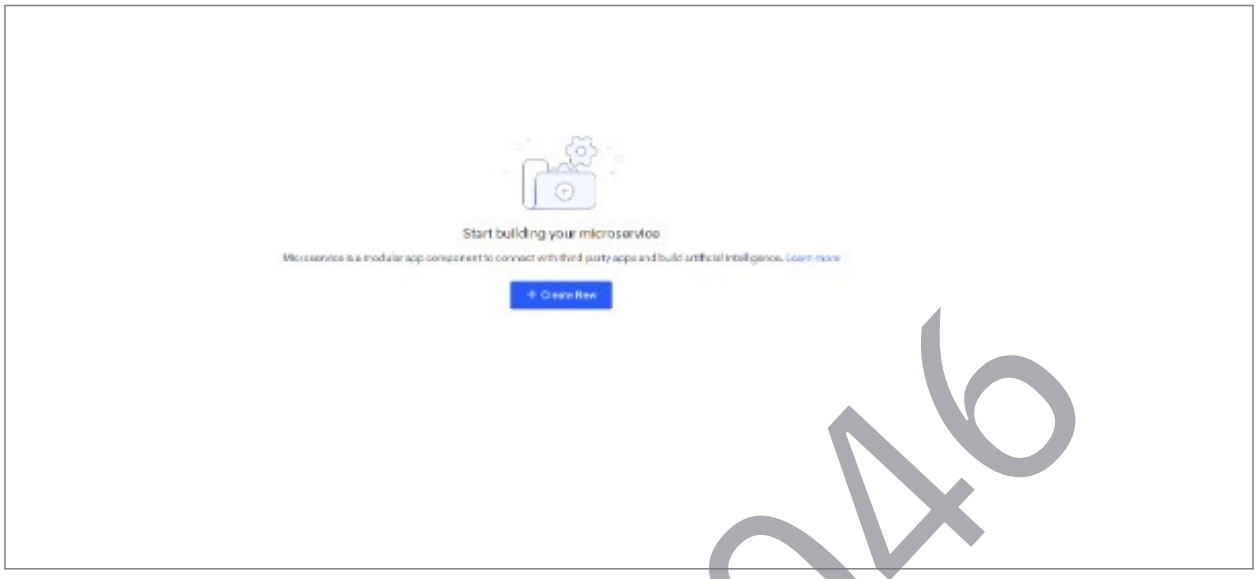
Step 7: Create the workflow — A new workflow was created on the library management form, with Record Event set to Created and Form Event set to Successful form submission.

Step 8: Write and save the Deluge script — A Deluge action was added. The script formats the preferred issue date (yyyyMMdd) and collection time (HHmmss) into a start value, fills the dateandtime and eventDetails maps, builds a description with the member name, email, book category, issue date and collection time, adds a 30-minute pop-up reminder, and calls zoho.calendar.createEvent with the calendar UID, the event map and the connection link name zoho_calendar_connection. The API response is shown using info, and the script was saved so it runs on every valid new submission.

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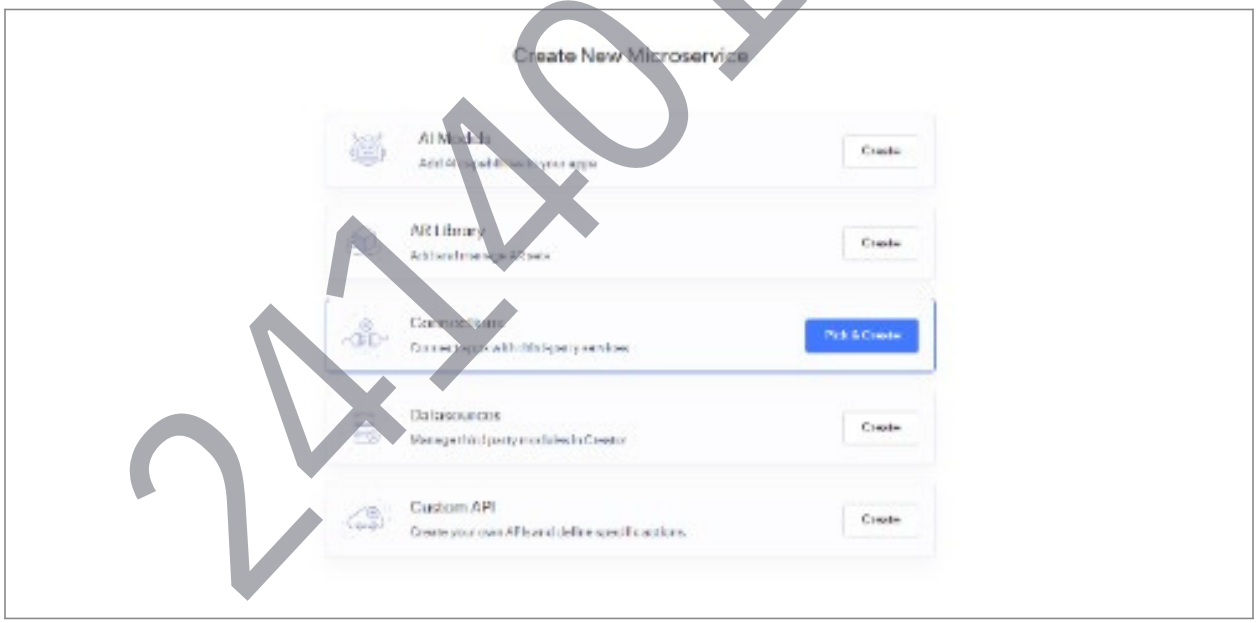
3. Output

3.1 Opening the Microservices section



The Microservices page is empty and offers Create New to start building a microservice for connecting with third-party apps.

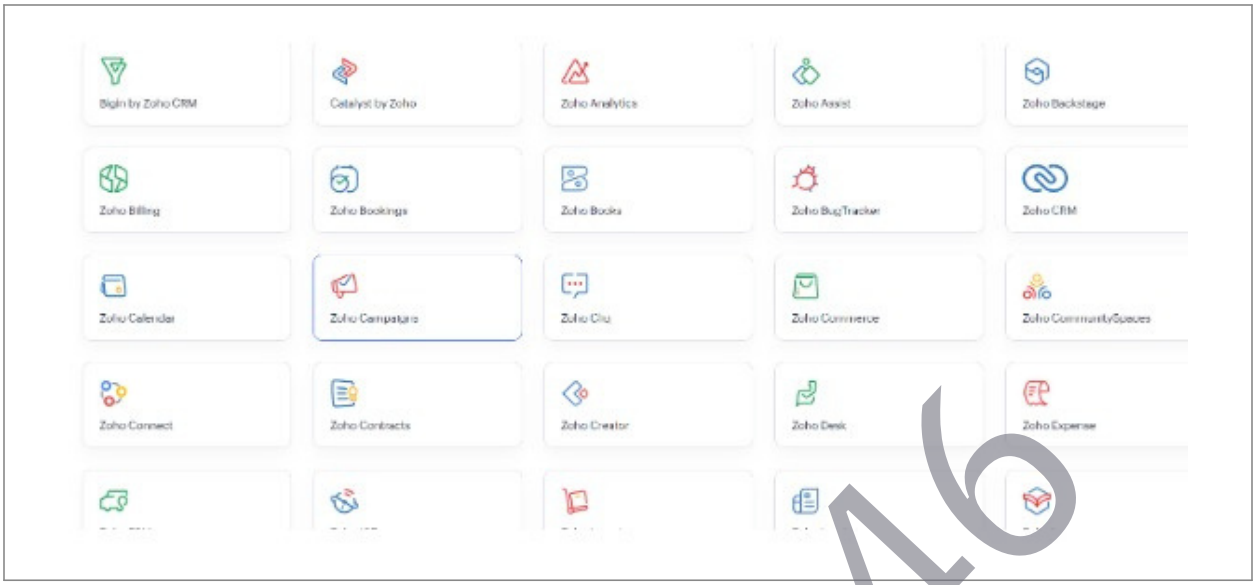
3.2 Choosing the Connections microservice



The Create New Microservice dialog lists AI Models, AR Library, Connections, Datasources and Custom API. Connections is highlighted because it links the application to an external Zoho service.

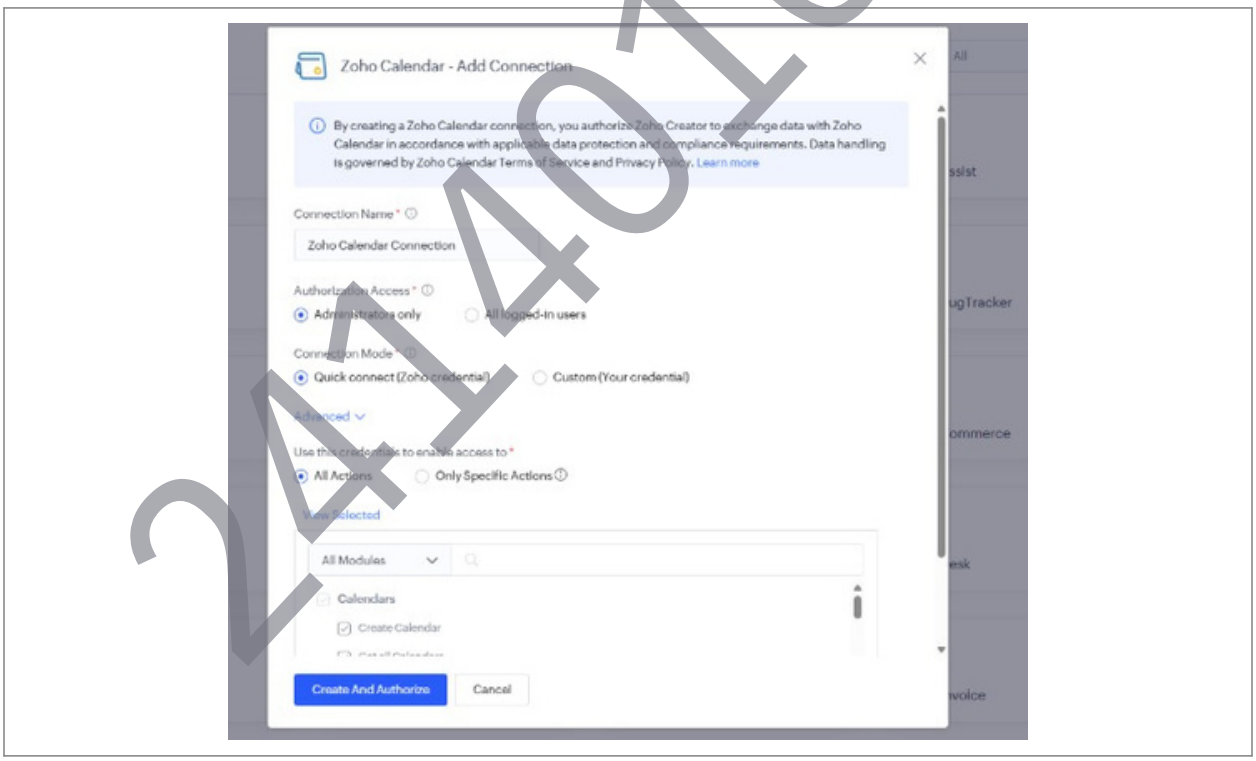
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3.3 Selecting the Zoho Calendar service



The service list shows the Zoho services a connection can be created for, among them Zoho Calendar, which is the service used for this integration.

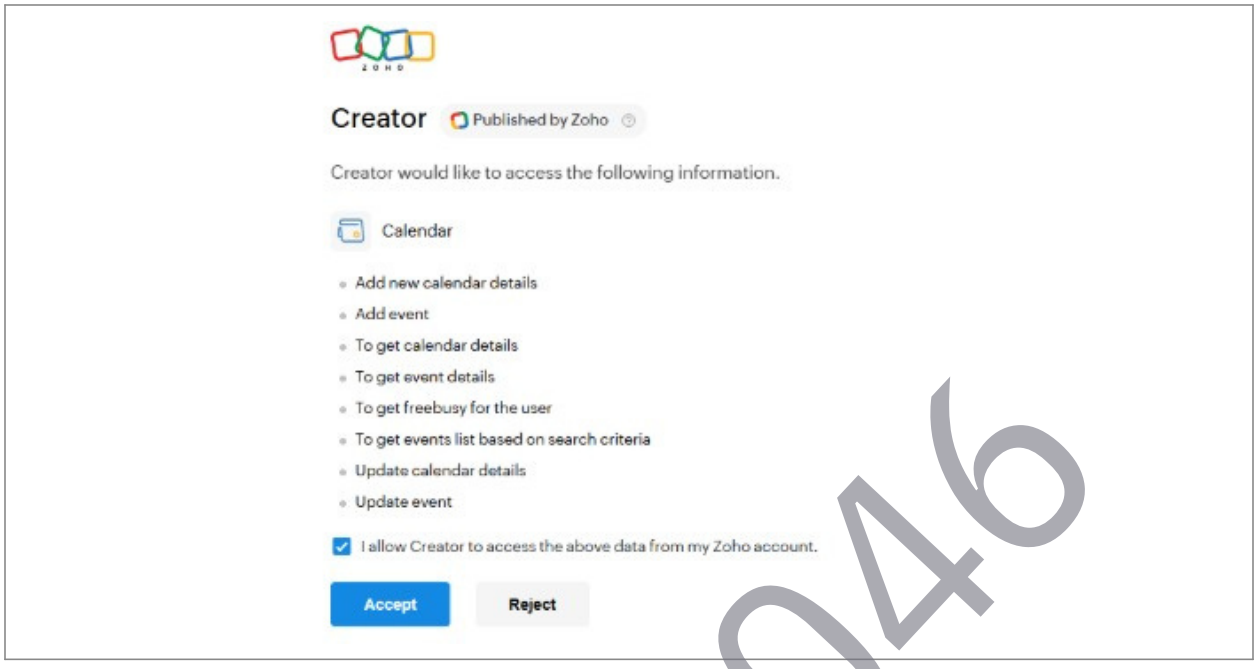
3.4 Adding the Zoho Calendar connection



The Add Connection form shows the connection name Zoho Calendar Connection, Administrators-only access, Quick connect mode and All Actions enabled, ready for Create And Authorize.

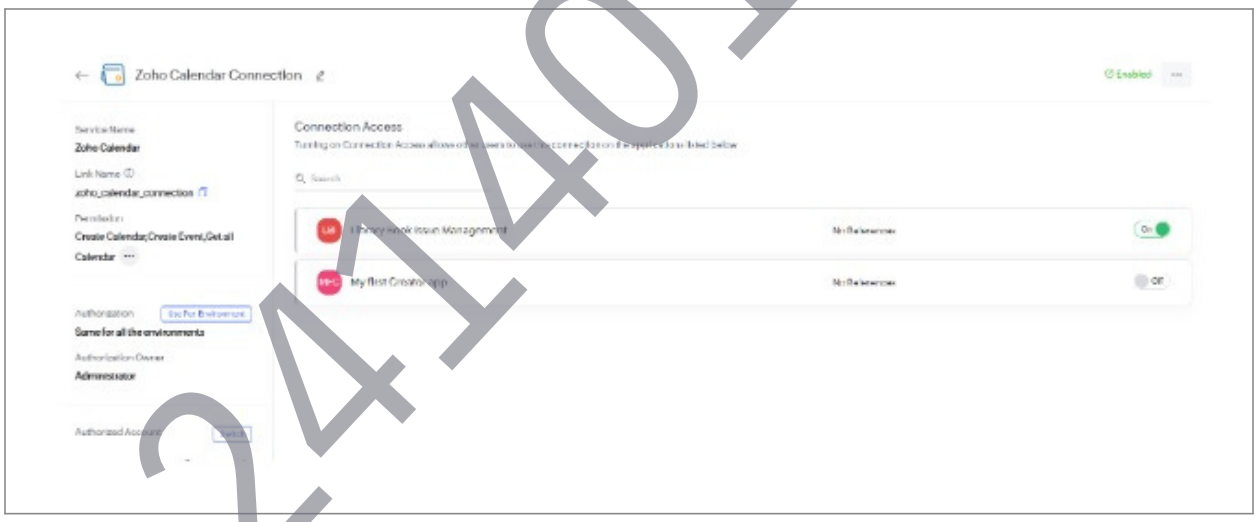
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3.5 Authorising Creator to access Zoho Calendar



The consent screen lists the Calendar operations Creator will be allowed to perform. Accepting it completes the authorisation of the connection.

3.6 Connection details and access



The saved connection shows Service Name Zoho Calendar, link name zoho_calendar_connection and permissions Create Calendar, Create Event and Get all Calendar, with the status Enabled. Access is On for Library Book Issue Management, so its Deluge scripts can use the connection, and Off for My first Creator app.

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3.7 Deluge script calling the Zoho Calendar API

```
startTime = input.Preferred_Issue_Date.toString("yyyyMMdd") + "T" + input.Preferred_Collection_Time.toString("HHmmss") + "Z";

dateAndTime.put("start",startTime);
dateAndTime.put("end",startTime);

eventDetails.put("dateandtime",dateAndTime);
eventDetails.put("isallday",false);

description = "Library Book Issue Request\n\n";
description = description + "Member: " + input.Name + "\n";
description = description + "Email: " + input.Email + "\n";
description = description + "Book Category: " + input.Book_category + "\n";
description = description + "Preferred Issue Date: " + input.Preferred_Issue_Date.toString() + "\n";
description = description + "Preferred Collection Time: " + input.Preferred_Collection_Time.toString() + "\n";

eventDetails.put("description",description);

reminder = Map();
reminder.put("action","popup");
reminder.put("minutes",30);

reminders = List();
reminders.add(reminder);

eventDetails.put("reminders",reminders);

calendarEvent = Map();
calendarEvent.put("eventdata",eventDetails);

response = zoho.calendar.createEvent(
    "YOUR_CALENDAR_UID",
    calendarEvent,
    "zoho_calendar_connection"
);

info response;
```

The Deluge editor shows the script. It converts the preferred issue date and collection time into the required start format, builds the dateandtime and event maps with a member description and a 30-minute pop-up reminder, and creates the event with zoho.calendar.createEvent using the connection zoho_calendar_connection.

4. Final Result

A connection to Zoho Calendar was created, authorised and enabled for the Library Book Issue Management application, and a Deluge script that uses it through the Zoho Calendar API was written for the library management form.

On each successful submission, the script builds the event start time from the preferred issue date and collection time, adds the member name, email, book category, issue date and collection time as the event description, sets a 30-minute pop-up reminder and sends the event to Zoho Calendar without manual entry by the library staff.

The exercise met its objective: a Zoho Creator application was integrated with another Zoho service using connections and an API call from Deluge, so book issue request data can flow automatically into the calendar.

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